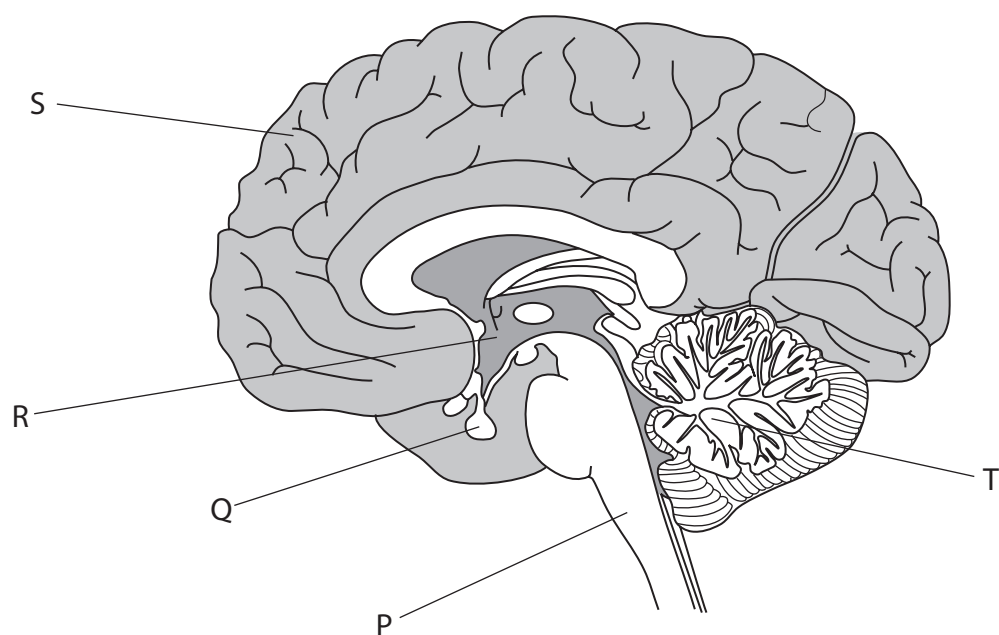


2 The human brain controls many functions.

(a) The diagram shows a section through the human brain.



(i) Which labelled part of the brain contains the cardiovascular control centre? (1)

- ☐ **A** P
- ☐ **B** Q
- ☐ **C** R
- ☐ **D** S

(ii) Which labelled part of the brain would control balance and coordinate movement? (1)

- ☐ **A** P
- ☐ **B** Q
- ☐ **C** R
- ☐ **D** T

(iii) Which labelled parts would be involved in controlling the solute concentration of blood plasma? (1)

- ☐ **A** P and Q
- ☐ **B** Q and R
- ☐ **C** P and R
- ☐ **D** S and T

(b) Dopamine and serotonin are chemicals produced in the brain that are essential for good health.

(i) Which of the following statements about dopamine is correct? (1)

- ☐ **A** dopamine can pass from the bloodstream into neurones in the brain
- ☐ **B** dopamine is a neurotransmitter
- ☐ **C** high levels of dopamine are linked to depression
- ☐ **D** low levels of dopamine cause the breakdown of myelin

- (ii) Drugs called selective serotonin re-uptake inhibitors (SSRIs) affect the reabsorption of serotonin at synapses in the brain.

This may help to reduce depression.

Explain why SSRIs can be used to treat depression.

(3)

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(Total for Question 2 = 7 marks)